

SUDHARSHAN RAM JAYARAMAN

COMPUTATIONAL SCIENCE AND ENGINEERING STUDENT

PROFILE

Motivated Master's student in Computational Science and Engineering at the Technical University of Munich with a strong foundation in Data driven analysis, Visualization techniques, Machine learning and numerical modeling. Experienced in numerical data analysis, 3D-CAD modeling, prototyping, and MATLAB-based simulations.

WORK EXPERIENCE

● Hilti Entwicklungsgesellschaft GmbH

Working student - Data Analyst

May, 2025-Present

- Consolidated NFC-enabled burnt battery data, sensor measurements, electronic logs, and CSV files from SharePoint by designing an end-to-end Microsoft Fabric data pipeline. Implemented data pipeline into a Fabric Lakehouse, performed cleaning, preprocessing, and feature engineering using Python notebooks (Pandas, NumPy), and stored curated datasets in downstream **SQL-based Lakehouses** for analytics.
- Applied **DBSCAN clustering** to detect voltage anomalies, conducted Pearson correlation analysis, and developed predictive models using Scikit-learn to support maintenance insights. Streamlined reporting with automated Excel workflows and supported agile delivery through **Jira (Kanban)** and **Microsoft Fabric**.

● Tata Technologies

Design Engineer

June, 2022-October, 2023

- Graphical representation of bench marked data of caliper and disc for EX30 model.
- Hands-on experience with **Agile methodology** in significant projects like Volvo XC90, XC60 and EX30.
- Recognized for significant contributions to **Pad Wear Indicator (PWI)** project, leading to an **18% profit** increase.

PROJECTS

● Machine learning in crowd Modelling - Semester project in Machine learning

Integrated machine learning and dynamical systems for simulating crowd behavior using Vadere, extended with epidemiological (SIR) dynamics. Applied ML principles such as **dimensionality reduction** (PCA, Diffusion Maps), **latent-space modeling** (VAEs), and **data-driven dynamical system** reconstruction through function approximation and **time-delay embedding** to analyze and visualize emergent crowd motion, with version control and collaboration managed via GitHub.

● Visual data analysis of sales trend in US market - Semester project in visual data analytics

Understanding sales trend in US with visual representation. Effect of visual channels in human perception is analyzed. Clustering, trend line and degree of perception were important parameter of study. Dashboard involving multiple plots and geographical represented along with filters in region, period were presented in tableau.

● Parallel CFD Solver Development - Semester Project in Scientific Computing

Developed a 2D CFD solver from scratch to simulate fluid dynamics using Navier-Stokes equations and understanding the software development cycle. Implemented numerical schemes on a staggered grid using finite differences. Parallelization was achieved using **MPI with domain decomposition** and GPU acceleration. Final solution was visualized using ParaView with velocity, pressure, and temperature fields. The project was extended further to **multigrids**, implementing V-cycle and W-cycle with version control via GitHub.

● Multi-Agent E-commerce Automation

Developed a multi-agent AI solution to manage customer queries, product recommendations, and order assistance across the e-commerce lifecycle. Built agent orchestration and pipeline logic to coordinate specialized agents and improved response relevance and task completion. Integrated data retrieval and context management to support personalized recommendations.

CERTIFICATIONS

- 100 days of code - Python bootcamp from Udemy
Dataframes, String manipulation, Oops, Regex, Data cleanup
- Applied Machine learning in Python - Coursera
KNN, Logistic regression, Multiclass classification, Naive Bayes, Random forest
- Prompt Engineering - Coursera
Data Analysis with LLM, Generative AI
- Databricks Fundamentals Accreditation - Databricks
- Spark, Hadoop and snowflake for Data engineering - Coursera

Gender: Male

Nationality: Indian

CONTACT

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<https://github.com/Sudharshan07709>

<https://my-portfolio-three-phi-40.vercel.app/>

EDUCATION

2024-Present

TECHNICAL UNIVERSITY OF
MUNICH

- Master in Computational Science and Engineering
- Coursework: Deep learning, Machine learning, Parallel programming, and Visual data analytics

2018-2022

AMRITA VISHWA VIDYAPEETHAM

- Bachelor in Mechanical Engineering
- CGPA: 9.3/10
- Relevant Coursework: FEM, Mechatronics, CFD, Thermal engineering, and Thermodynamics

TECHNICAL SKILLS

- Simulation tools- ANSYS Workbench, Fluent
- Programming skills- C++, MATLAB, Python(Pytorch, numpy, pandas, Sci-kit, matplotlib), Latex
- Visualization tools- Tableau, Viz3D, Excel
- Design and PLM tools- Catia, Fusion 360, Creo, Teamcenter
- Management tools- JIRA, sharepoint, SQL, Fabric, Confluence
- other tools - n8n, Dataiku

LANGUAGES

- English (C1)
- Tamil (Native)
- German (B2)